TransFusion-CR: Two-Phase SAR-to-Optical Translation and Deep Feature Fusion for Cloud Removal

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***Abstract*— Multispectral image (MSI) is essential for remote sensing (RS) tasks. Unfortunately, the presence of clouds will lead to a reduction in the accuracy and availability of ground information. Synthetic aperture radar (SAR) is unaffected by cloud cover, which can serve as a supplemental information source to alleviate the difficulty of cloud removal in MSI. However, SAR images suffer from the problem of speckle noise and contain information only in a single spatial dimension. To achieve cloud removal effectively by fusing MSI and SAR, we propose the TransFusion-CR in this article. It is a novel two-phase SAR and MSI translation-fusion cloud removal network and designed to enhance the information gains provided by SAR. We use a convolutional neural network (CNN) with a large kernel and design special modules to process the global and local information so that TransFusion-CR can reconstruct clear spatial details and accurate spectral information without artifacts. In the translation phase, we establish an SAR frequency extraction (SFE) module based on the discrete wavelet transform (DWT) to separate and process different frequency information of the SAR image. This assists in dealing with the high-frequency speckle noise while generating translated cloud-free MSI with clear spatial information and reliable spectral information. Subsequently, in the fusion phase, a specially designed feature fusion module realizes a weighted triple-source image fusion guided by the cloud mask score (CMS) image, which high-quality cloud-free MSI. Finally, a new loss function is proposed to balance the learning focus and achieve joint optimization. Extensive experiments demonstrate that TransFusion-CR outperforms the state-of-the-art by about 1.3 dB in terms of PSNR when the percentage of thick cloud cover is more than 60% and can construct reliable terrestrial information even in the complete absence of surface data.**

***Index Terms*— Cloud removal, data fusion, deep learning, multispectral image (MSI), synthetic aperture radar (SAR).**

I. Introduction

EARTH observation is essential for comprehending the world, production, and creation. In recent years, satellite observations have provided remote sensing (RS) images that possess various characteristics, including multitemporal, multimodal, and multisensor [1]. However, the cloud-corrupted area in optical RS images remains one of the main challenges in downstream tasks such as target classification and ground monitoring. The loss of vital ground information such as water bodies, vegetation, and land details is caused by the thick cloud cover in these images. Consequently, removing clouds, especially thick clouds, is an essential preprocessing step in RS image processing.

Removing thick clouds in multispectral images (MSIs) is more complex than removing thin clouds due to the excessive lack of spatial, temporal, and spectral domain information. There are a certain amount of works in this field that can be categorized into four types based on the data source: spatial-based, spectral-based, temporal-based, and hybrid methods [2], [3]. Among them, multitemporal approaches have been demonstrated to be effective in addressing the challenge of thick cloud removal [4], [5], [6]. However, such approaches require a series of MSI images taken at different times (usually three or more), which is demanding in terms of data acquisition and processing. As a result, such methods struggle to deal with rapidly changing surfaces. In this regard, multisource data can provide complementary information to sole optical data and complete this task with lower data requirements. Synthetic aperture radar (SAR) image is a type of all-season RS data. Its unique imaging mechanism and strong penetration lead to image characteristics that are free of cloud cover and weather interference. SAR images provide the additional ground information that are missing in optical images, solving the source problem of the lack of information due to thick cloud cover. Therefore, the thick cloud removal task can be achieved by efficiently fusing SAR with MSI.

The current research can be divided into two categories according to the number of fusion phases: direct fusion [7] and two-phase fusion [8]. Direct fusion uses SAR and cloudy MSI to extract feature maps for fusion; two-phase fusion first completes translating SAR inputs into MSI outputs and then fuses them with cloudy MSI. Due to the distinct difference between the SAR and MSI, the information available from the SAR image with a single spatial dimension in direct fusion is insufficient compared to the information available from MSI with a dozen spectral bands. Nevertheless, the SAR image can provide more information to the image fusion after being translated into synthetic MSI. Meanwhile, phase-by-phase processing provides a large space for control and optimization of the training network.

However, existing two-phase fusion approaches still exhibit limitations and challenges due to the following issues.

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1) In the phase of SAR to MSI translation, most methods employ generative adversarial networks (GANs) [9], [10]. Due to the inherent framework and adversarial learning mechanism of GAN, the predicted image cannot fully recover the exact content, which leads to the inability of GAN methods to meet the fidelity requirement. Some other attempts adopt the transformer architecture [7], which has a global attention design and can provide guidance for local detail recovery in cloud removal. However, the computation complexity of the transformer is quadratically correlated with the input size, which can cause problems in high-resolution image reconstruction.
2) Most existing SAR-to-optical methods focus on the task of converting SAR images to RGB images (only have three spectral channels). For MSI with a dozen channels, it contains more feature information than SAR image with only a single channel. Therefore, although the existing methods can generate an ideal translated RGB image, it is still difficult to generate an effective translated MSI. This makes the translated MSI unable to provide effective feature information for the subsequent fusion phase.

To address the above issues, we propose a novel two-phase fusion cloud removal network (TransFusion-CR) for image translation-fusion using SAR and MSI in this work. Considering the low fidelity of GAN and the high computational load of the transformer, we choose ConvNeXt [11], a convolutional neural network (CNN), as the backbone. By improving key components in ResNet with reference to the design of the structural parameters of the transformer, ConvNeXt is proved to be comparable to the transformer in a variety of image processing tasks. Based on this backbone, we further design an end-to-end neural network to implement two-phase feature learning. In the first translation phase, we aim to optimize the effect of SAR-to-optical translation and equip translated MSI with more comprehensive and useful information. To achieve this, we construct an SAR frequency extraction (SFE) module based on the discrete wavelet transform (DWT) to separate the high- and low-frequency features. The features are then applied to feature extraction of the encoder and image reconstruction of the decoder, respectively. In the second fusion phase, we perform a weighted triple-source image fusion to produce high-quality cloud-free MSI. To this end, we first compute a cloud mask score (CMS) image to provide reliable information about the cloud region, which also shows the thickness of the cloud. Furthermore, we establish a cloud mask score-guided feature fusion (CFF) module, which uses the feature weight map extracted from the CMS image as a guide to calculate the weight distribution for the fusion of translated MSI and cloudy MSI. Finally, we propose a specially designed loss function for joint optimization of the proposed two-phase fusion network, which also applies the CMS image to provide the appropriate weight of loss in different phases.

In summary, the main contributions of this work are as follows.

1) We introduce a novel end-to-end two-phase translation-fusion cloud removal network, named TransFusion-CR, for cloud removal tasks based on the fusion of SAR and MSI. We use a large kernel of ConvNeXt block in the convolution layer to obtain the global information and design special modules to process the local information. Therefore, TransFusion-CR can take into account both feature translation and fusion so that the reconstruction result has clear spatial details and accurate spectral information.
2) Technically, we introduce three novel designs: the SFE module, the CFF module, and a new loss function. SFE processes high- and low-frequency information of SAR, while CFF achieves guided triple-source image fusion. These two specially designed modules can efficiently extract and fuse the local information, allowing the network to reconstruct the ground details of the cloudy area while maintaining the clarity of the cloud-free area. Furthermore, a well-designed loss function is proposed to balance the learning focus and achieve joint optimization.
3) Extensive experiments on a real dataset have been carried out, and the experimental results show that TransFusion-CR with a two-phase strategy can achieve excellent effects for cloud removal, especially for thick cloudy MSI.

The rest of this article is organized as follows. After the introduction in Section I, the related works in this research area are presented in Section II. The proposed method is elaborated in Section III, including the two-phase neural network architecture and the loss function. Section IV describes the dataset, experimental settings, and the results obtained, and conducts ablation experiments to demonstrate the effectiveness of the proposed method. Section V further discusses the results and concludes our work.

II. Related Work

A. Single-Modal Deep Learning

Deep learning has contributed significantly to cloud removal in RS images, especially since the emergence of CNNs. Neural networks can effectively extract abstract features from images and exhibit high adjustability and strong generalizability. Consequently, CNN has rapidly advanced in cloud removal for satellite images, leading to the development of numerous algorithm variants [12], [13], [14]. In addition to classical CNNs, the introduction of GAN by Goodfellow et al. [15] has made substantial contributions to research progress in this field. Nevertheless, with the limitation of the single modal in optical images, the effectiveness of thick cloud removal is impeded because of excessive information loss. Therefore, single-modal-based deep learning algorithms that rely solely on a single image mainly aim at removing thin clouds. SpA-GAN [16], a GAN-based model equipped with the spatial attention mechanism, was proposed to remove thin clouds utilizing the RICE dataset. Li et al. [17] employed a semi-supervised method named CR-GAN-PM, using GANs and a physical model of cloud distortion to remove thin clouds. Another method named CycleGAN [18] overcame the rigorous constraints of pairing cloudy and cloud-free

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images in the training data to remove the cloud. Due to the challenge of thick cloud removal by using only a single modal and a single image, applying a series of temporal images of identical scenes has become a potent technique in this field. Studies [19], [20] integrate multiple temporal images for reconstructing the cloud-corrupted regions. In addition, Zheng et al. [21] utilized highly correlated spatial–spectral-time information contained in multitemporal RS images to develop SSTC-CR, a semi-blind thick cloud removal approach, to quickly explore complex sea temperature relationships in the clouded region. It can be observed that for single-modal cloud removal tasks, information sources typically comprise of spatial complementary information in temporal-series imagery and band complementary information from multispectral imagery.

B. Multimodal Deep Learning

1) Direct Fusion: Removing clouds from RS images with single-modal deep learning can be challenging due to the limited information available. However, the increasing availability of multimodal data in recent years has provided a new perspective for solving this problem. Sentinel-2 SAR images can penetrate clouds to obtain surface information, providing guidance for reconstructing areas covered by clouds. Research on cloud removal using direct fusion of SAR and optical images has rapidly developed, benefiting from the contributions of deep learning networks in reconstructing missing information.

Since GANs were proposed, GANs and their derived variants have been widely used in direct SAR and optical image fusion for cloud removal. Grohnfeldt et al. [22] designed a cGAN to generate cloud-free images from cloudy optical images with the help of SAR images. Similarly, other researchers have also used cGANs to recover images. SAR2OPT [9] utilizes pairs of SAR-optical patches to train a network of cGANs and reconstructs the other spectral bands in the image relying only on the SAR data. While GANs are highly capable of generating images, their predictions can be unstable when trained on a wide range of corrupted input data. In addition, the level of detail in GAN predictions may not be accurate enough to reflect ground-truth situations, as their predictions are only to deceive the discriminator. This can result in unsatisfactory generated results. To enhance robustness, Meraner et al. [23] proposed the DSen2-CR network, a deep residual neural network inspired by ResNet. This greatly improves the stability of prediction. However, since the residual cascade can only obtain incomplete information from the SAR image, the prediction effect in a specific area may be reduced. To enhance the benefits of SAR information in image reconstruction, the GLF-CR [7] integrates local–global information and uses a two-stream network with two special modules. This approach recovers texture details and maintains the information structure of cloud-free regions.

2) Two-Phase Fusion: In parallel with the direct fusion approach, there are also other existing methods [8], [24], [25] that utilize a two-phase fusion approach. These methods transform SAR images into synthetic MSI in the first phase and then fuse the synthetic MSI with the cloudy MSI to reconstruct the cloudy area information in the second phase. Two modules are involved in this two-phase fusion process: one for translation and the other for fusion. Specifically, Gao et al. [25] designed a GAN-based network for cloud removal using simulated clouds first and applied it to real cloud removal in optical images. Darbaghshahi et al. [24] utilized a dual-GAN strategy with dilated residual inception blocks for SAR-to-optical image translation and cloud removal, while Zhang et al. [8] developed an end-to-end cloud removal network that integrates SAR-to-optical image translation with an attention mechanism.

In conclusion, the current two-phase fusion is mostly based on GAN, which still faces the same issues as direct fusion, such as unstable predictions and inaccurate details. Furthermore, in the first phase of translation, the current method only converts the SAR image into a three-channel synthetic optical image (RGB image) [24], which provides limited information and cannot achieve the optimal solution for information utilization in removing clouds from MSI.

III. Methodology

In this section, we introduce the end-to-end two-phase network architecture of TransFusion-CR in detail. First, the overview of the network is presented in Section III-A. The specific design of the translation phase and fusion phase, including the SFE and CFF modules, are presented in Sections III-B and III-C, respectively. Finally, the CMS-guided loss designed to optimize the training process is given in Section III-D.

A. Overview

The network architecture of the proposed TransFusion-CR is illustrated in Fig. 1(a). The end-to-end TransFusion-CR network consists of two phases, responsible for translation and fusion, respectively. We choose the ConvNeXt (tiny) [11] as the backbone of the main layers because of its accuracy, robustness, and scalability. Furthermore, an SFE module, a CFF module, and a loss function are innovatively constructed for the two-phase network.

Inspired by [26], who introduces DWT into the neural network for dehazing, we construct the SFE module (see Section III-B) to separatively extract the high- and low-frequency feature of SAR image in the SAR-to-optical translation phase. In each layer, the low-frequency feature of SAR is transmitted to the next layer to mine the deep structural information. Meanwhile, the high-frequency feature of SAR is transmitted to the decoder through the skip connection, so that the texture detail of the SAR image can be effectively restored to the translated MSI. We utilize $S_{\text{sar}} \in \mathbb{R}^{H\times W\times C_{\text{sar}}}$ to denote the input image, where $H \times W$ is the size of SAR image, and C_{sar} is the number of SAR's channel. The translated cloud-free MSI $\hat{S}_{\text{opt}} \in \mathbb{R}^{H\times W\times C_{\text{msi}}}$ can be obtained by

$$\hat{S}_{\text{opt}} = \mathbf{F}_{\text{translation}}(S_{\text{sar}}) \tag{1}$$

where $\mathbf{F}_{\text{translation}}(\cdot)$ denotes the SAR-to-optical translation phase and C_{msi} is the number of cloudy MSI's channel.

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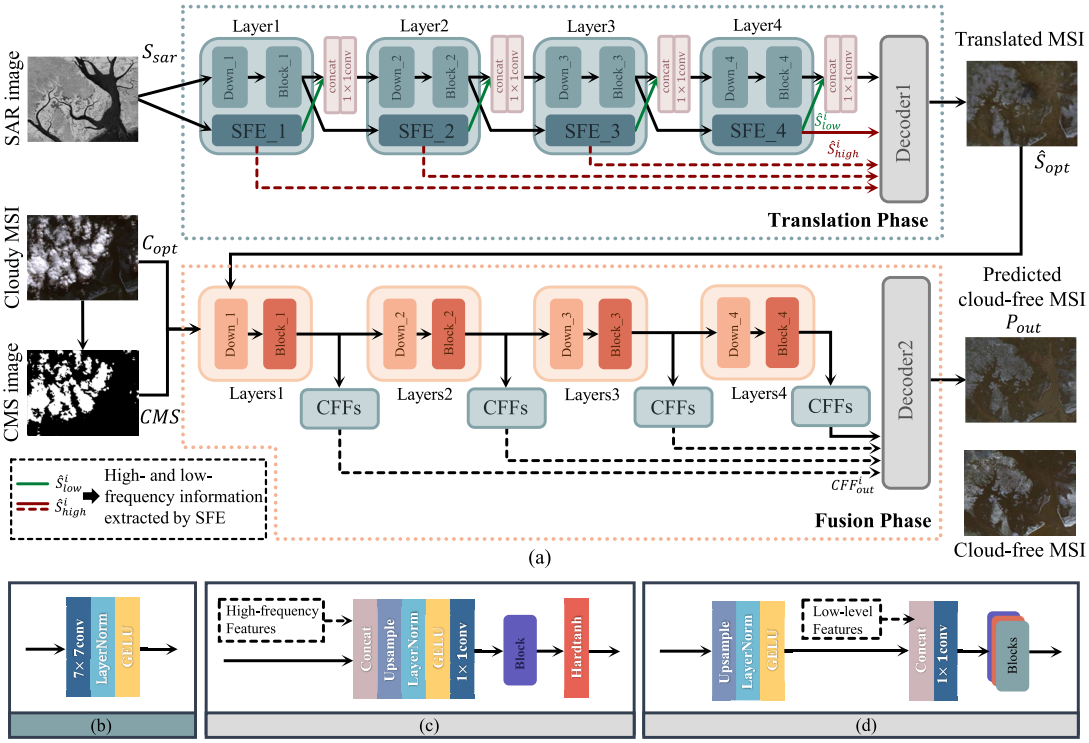

Fig. 1. Overview of the TransFusion-CR network. It is an end-to-end network composed of two phases: one for SAR-to-optical translation and the other for image fusion. In the first translation phase, the SFE module extracts deep high- and low-frequency features from the SAR to enhance its gain when transmitted to the next phase. In the second fusion phase, the CFF module accomplishes weighted triple-source fusion. The detailed structure of the SFE and CFF modules is illustrated in Figs. 2 and 3, respectively. (a) TransFusion-CR network architecture. (b) Block. (c) Decoder1. (d) Decoder2.

In the fusion phase, we construct the CFF module (see Section III-C) for triple-source fusion. We first calculate the probability of each pixel being the cloud in the cloudy MSI based on the cloud mask estimation method employed by [23]. Then, we keep the score of pixels with probability greater than 0.5 and set the pixel with probability less than 0.5 to 0 to get our CMS image CMS $\in \mathbb{R}^{H\times W\times 1}$. Furthermore, in the CFF module of each layer, we fuse the features of translated MSI and cloudy MSI through the spatial attention mechanism guided by the feature of CMS and then transmit the fused feature to the decoder by skip connection to restore the predicted cloud-free MSI. We utilize $C_{\text{opt}} \in \mathbb{R}^{H\times W\times C_{\text{msi}}}$ to denote the cloudy MSI. And the predicted cloud-free MSI P_{out} can be obtained by

$$P_{\text{out}} = \mathbf{F}_{\text{fusion}}\left(\hat{S}_{\text{opt}}, C_{\text{opt}}, \text{CMS}\right) \tag{2}$$

where $\mathbf{F}_{\text{fusion}}(\cdot)$ denotes the fusion phase and $P_{\text{out}} \in \mathbb{R}^{H\times W\times C_{\text{msi}}}$ is the final prediction of cloud-free MSI.

In our TransFusion-CR, the composition of Layer1 to Layer4 in the translation phase includes the downsample layers (Down_i), the ConvNeXt blocks (Blocks_i), and the SFE (SFE_i), where $i = 1, 2, 3, 4$. Specifically, the first layer of the downsampling module (Down_1) performs a 4×4 nonoverlapping convolution, while the other three layers (Down_2, Down_3, and Down_4) perform a 2×2 nonoverlapping convolution. The four downsampling layers have output channels of (96, 192, 384, and 768). The ConvNeXt Block [as shown in Fig. 1(b)] employs a 7×7 convolutional layer followed by layer normalization with GeLU activation. The larger kernel of the convolutional layer enables the network to capture global image features better. In each layer, the SFE module extracts high- and low-frequency information from the input, i.e., $\hat{S}^i_{\text{high}}$ and $\hat{S}^i_{\text{low}}$, respectively. Then, $\hat{S}^i_{\text{low}}$ and the output of the block B^i_{out} are sent to the next layer for downsampling, while $\hat{S}^i_{\text{high}}$ jumps into the Decoder1 after the concatenation [as shown in Fig. 1(c)] to be upsampled. In the fusion phase, Layer1 to Layer4 follow a similar structure design to that of the former phase. Each contains the same three minor layers as those in the translation phase, and the three inputs, i.e., $\hat{S}_{\text{opt}}$, C_{opt} and CMS, go into these three layers for synchronized downsampling to extract features. The output of each layer is fused through CFFs and the outputs of CFFs ($\text{CFF}^i_{\text{out}}$) are input to Decoder2 [as shown in Fig. 1(d)] for upsampling decoding. The effectiveness of our network design is later verified in the ablation study (see Section IV-F).

B. SFE: SAR Frequency Extraction

In our work, the SFE module is proposed for the first time to process different frequency information in the SAR image.

The 2-D DWT applies two sets of filters to process the input sequentially. One set is applied horizontally to the rows and the other is applied vertically to the columns, creating four filters: one low-pass filter $\mathbf{f}_{\text{LL}}$ to achieve the image approximation and three high-pass filters $\mathbf{f}_{\text{LH}}$, $\mathbf{f}_{\text{HL}}$, and $\mathbf{f}_{\text{HH}}$ to achieve edge extraction. These filters are equivalent to a convolution operation with fixed parameters and stride [26]

$$\mathbf{f}_{\text{LL}} = \begin{pmatrix} +1 & +1 \\ +1 & +1 \end{pmatrix}, \quad \mathbf{f}_{\text{LH}} = \begin{pmatrix} -1 & -1 \\ +1 & +1 \end{pmatrix}$$
$$\mathbf{f}_{\text{HL}} = \begin{pmatrix} -1 & +1 \\ -1 & +1 \end{pmatrix}, \quad \mathbf{f}_{\text{HH}} = \begin{pmatrix} +1 & -1 \\ -1 & +1 \end{pmatrix}. \tag{3}$$

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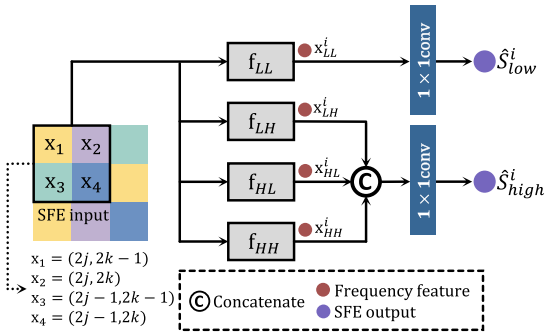

Fig. 2. Detailed structural design of the SFE module. The SFE input (S_{sar}) is transformed into four subbands by four filters defined in (3). After concatenation and full connection, the low- and high-frequency information ($\hat{S}^i_{low}$ and $\hat{S}^i_{high}$) enters the next layer and Decoder1, respectively.

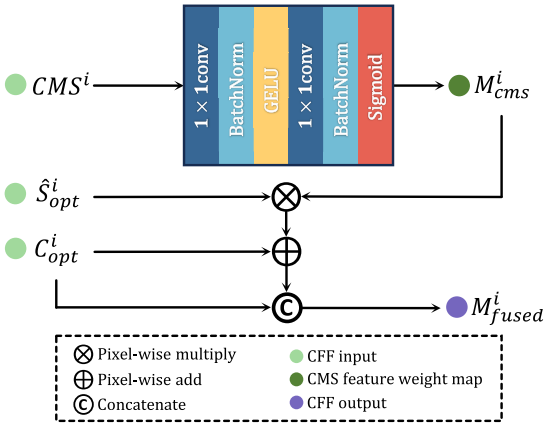

Fig. 3. Detailed structural design of the CFF module. The CMS feature weight map is obtained from CMS after a series of convolution, normalization, and activation. The final fusion is accomplished according to (5).

After the transform process, the input $\mathbf{x}^i$ becomes four subband parts, $\mathbf{x}^i_{\mathrm{LL}}$, $\mathbf{x}^i_{\mathrm{LH}}$, $\mathbf{x}^i_{\mathrm{HL}}$, and $\mathbf{x}^i_{\mathrm{HH}}$, where $\mathbf{x}^i_{\mathrm{LL}}$ is the low-frequency information of the SAR image and the other three are the high-frequency information. Specifically, $\mathbf{x}^i_{\mathrm{LL}}$ is obtained from (4), and the other three subbands can be expressed similar to (4) according to the expressions of the high-pass filters [as shown in (3)]

$$\mathbf{x}^i_{\mathrm{LL}} = \mathbf{x}^i(2j-1, 2k-1) + \mathbf{x}^i(2j-1, 2k) + \mathbf{x}^i(2j, 2k-1) + \mathbf{x}^i(2j, 2k) \quad (4)$$

where $\mathbf{x}^i(j,k)$ denotes the index of pixels in the SAR image.

The structure diagram of the SFE module is shown in Fig. 2. After completing the feature extraction of the four subbands, the three new high-frequency information ($\mathbf{x}^i_{\mathrm{LH}}$, $\mathbf{x}^i_{\mathrm{HL}}$, and $\mathbf{x}^i_{\mathrm{HH}}$) are concatenated and fully connected as the overall high-frequency feature $\hat{S}^i_{\mathrm{high}}$ (i.e., the details of the land texture contained in the SAR image) and $\hat{S}^i_{\mathrm{high}}$ jumps into the decoder. The low-frequency information ($\mathbf{x}^i_{\mathrm{LL}}$) is fully connected as the low-frequency feature $\hat{S}^i_{\mathrm{low}}$ (i.e., the approximation of the SAR image) to proceed to the next layer for downsampling.

C. CFF: CMS-Guided Feature Fusion

To enhance feature fusion quality, we further construct the CFF module to further exploit information in cloud-free and cloudy areas.

After the SAR-to-optical translation phase, the cloud-free translated MSI obtained from SAR ($\hat{S}_{\mathrm{opt}}$) serves as the input for this fusion phase, providing supplementary information. $\hat{S}_{\mathrm{opt}}$, along with C_{opt} and CMS, are fed into the fusion phase for training.

The structural design of the CFF module is illustrated in Fig. 3. When the CMS enters this module, it performs convolution, batch normalization, and activation processing, resulting in a CMS feature weight map M^i_{cms}. This weight map is then pixel-wise multiplied with the cloud-free translated MSI, modulating the information provided by $\hat{S}^i_{\mathrm{opt}}$ in the cloudy areas. The multiplied output is added to C^i_{opt}. Then, the output feature map M^i_{MidFused} can be obtained by

$$M^i_{\mathrm{MidFused}} = M^i_{\mathrm{cms}} \otimes \hat{S}^i_{\mathrm{opt}} + C^i_{\mathrm{opt}} \quad (5)$$

where $\otimes$ denotes the pixel-wise multiplication.

M^i_{MidFused} is then concatenated with the cloudy MSI to obtain the final single feature map as the output of the CFF module, i.e., M^i_{fused}.

D. Loss Function

We design a new loss function $\mathcal{L}_{\mathrm{loss}}$ for the TransFusion-CR network to optimize training. This loss function is specially designed for the two-phase learning strategy, as illustrated in (6). It is comprised of two main components: a translated cloud-free loss ($\mathcal{L}_{\hat{S}_{\mathrm{opt}}}$) and a prediction loss ($\mathcal{L}_{P_{\mathrm{out}}}$)

$$\mathcal{L}_{\mathrm{loss}} = \mathcal{L}_{\hat{S}_{\mathrm{opt}}} + \mathcal{L}_{P_{\mathrm{out}}}. \quad (6)$$

In $\mathcal{L}_{\hat{S}_{\mathrm{opt}}}$, we calculate the difference between the ground truth for cloud-free MSI and $\hat{S}_{\mathrm{opt}}$ based on the Smooth $L1$ loss, and utilize CMS as the guidance of weight distribution to increase the weight of cloudy areas. Correspondingly, we construct $\mathcal{L}_{P_{\mathrm{out}}}$ to evaluate the differences in P_{out} and utilize $1-\mathrm{CMS}$ as the guidance of weight distribution to increase the weight of cloud-free areas. $\mathcal{L}_{\hat{S}_{\mathrm{opt}}}$ and $\mathcal{L}_{P_{\mathrm{out}}}$ are calculated as follows:

$$\mathcal{L}_{\hat{S}_{\mathrm{opt}}} = \frac{\mathcal{L}1(\hat{S}_{\mathrm{opt}}) + \mathcal{L}1(\hat{S}_{\mathrm{opt}}) \cdot \mathrm{CMS}}{N_{\mathrm{ttp}}} \quad (7)$$

$$\mathcal{L}_{P_{\mathrm{out}}} = \frac{\mathcal{L}1(P_{\mathrm{out}}) + \mathcal{L}1(P_{\mathrm{out}}) \cdot (1-\mathrm{CMS})}{N_{\mathrm{ttp}}} \quad (8)$$

where $\mathcal{L}1(\cdot)$ denotes the smooth $L1$ loss and $N_{\mathrm{ttp}} = H \times W \times C_{\mathrm{msi}}$ is the number of total pixels, respectively.

IV. Experiments and Results

A. Dataset and Evaluation Metrics

1) Dataset: The experiments are conducted on the large-scale RS dataset SEN12MS-CR [27] built upon the openly available dataset SEN12MS [28]. SEN12MS is a collection of 180 662 samples from nonoverlapping regions of interest (ROIs), containing Sentinel-1 dual-pol SAR images and Sentinel-2 MSIs. The collection overcomes spatial limitations regarding coverage, diversity, and sample availability and provides data support for the gathering of SEN12MS-CR.

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TABLE I
QUANTITATIVE RESULTS COMPARED WITH THE STATE-OF-THE-ART METHODS

Method	**Input** (SAR/Optical/Both)	**PSNR** (dB) ↑	**SSIM** ↑	**SAM** (°) ↓
SpA GAN [16]	Optical	22.8273	0.7553	8.5148
SAR2OPT [9]	SAR	18.7645	0.7192	12.3260
SAR-Opt-cGAN [22]	Both	20.9992	0.8034	9.9521
DSen2-CR [23]	Both	25.2412	0.7886	5.2650
GLF-CR [7]	Both	24.5902	0.7667	7.4271
TransFusion-CR (Ours)	Both	**25.7649**	**0.8344**	**4.4000**

Based on the SEN12MS dataset, SEN12MS-CR comprises 122 218 samples from 169 nonoverlapping ROIs distributed evenly across all the continents and seasons, with each patch consisting of a triplet of Sentinel-2 cloudy and cloud-free 13-band MSIs, as well as the correspondent Sentinel-1 SAR image which includes VV and VH polarization information. Each image is 256 × 256 pixels in size, and each band of the SAR and optical images is rescaled to range $[0, 1]$. Specifically, the VV and VH polarizations of the SAR images are clipped to values of $[-25, 0]$ and $[-32.5, 0]$, respectively, and then normalized to range $[0, 1]$, while all bands of the MSIs are clipped to values $[0, 10\,000]$ first and normalized to range $[0, 1]$ as well. In addition, a median filter is applied to the SAR images to suppress speckle noise. To ensure the consistency of content, the data applied in the study is checked to exclude images with varying content due to different acquisition times.

Finally, we select two subsets of data from the original dataset named DatasetV1 and DatasetV2 due to limited arithmetic power. DatasetV1 contains 2822 images and 100 images from the first 5 ROIs under the folder of ROIs1158_spring as the training set and test set, respectively; DatasetV2 contains 2000 images and 100 images from 10 ROIs, which are used as the training set and test set, respectively. The distinction between the two datasets is that the former is used for all comparison methods to obtain the experimental results presented in Section IV-C. The latter dataset includes more landscape types, such as farmland and urban areas, resulting in a higher level of detail demand. As a result, it is compared to the optimal comparison method and further discussed in Section IV-D.

2) Metrics: Several metrics are applied to quantitatively evaluate the experimental results, including the peak signal-to-noise ratio (PSNR) in units of decibel (dB), the unitless structural similarity index measure (SSIM), and spectral angle mapper (SAM) in units of degrees (°). PSNR is an evaluation metric commonly used in pixel-wise image reconstruction to evaluate the quality of predicted images, such as fidelity of information, average error level, and so on. SSIM quantifies the degree of spatial structural similarity between two images based on visual perception principles, considering image brightness, contrast, and structure, while SAM focuses on comparing the degree of spectral similarity between two images in RS images.

B. Experimental Settings

The proposed network is implemented using PyTorch and trained on a single NVIDIA RTX 3090Ti GPU (24 GB) with a batch size of 16. The maximum epoch of training iterations is set to 1000. The AdamW optimizer is used with a learning rate of 0.001. To augment the data used in training, the raw input data with a size of 256 × 256 pixels is randomly cropped to 128 × 128 pixels in each iteration during the training process.

C. Results and Analysis

To evaluate the effectiveness of the proposed network in cloud removal, it is compared with several state-of-the-art cloud removal methods, including SpA GAN [16], SAR2OPT [9], SAR-Opt-cGAN [22], DSen2-CR [23], and GLF-CR [7]. SpA GAN uses GAN with the spatial attention mechanism to achieve thin cloud removal, applying only a single optical modal. Both SAR2OPT and SAR-Opt-cGAN are GAN-based networks with similar structures. SAR2OPT uses conditional GANs (cGANs) to learn the nonlinear mapping from SAR to optical images and then relies only on SAR input to translate to the optical output; SAR-Opt-cGAN is a cGAN built upon the pix2pix [10] framework. It employs a U-Net as the generator and a PatchGAN [29] as the discriminator, which uses SAR as prior information to guide the fusion of SAR and MSI. DSen2-CR derives from the DSen-2 ResNet and also utilizes SAR as a prior to guide the cloud removal process. GLF-CR combines the idea of the convolutional network and transformer to create a two-stream network for fusing MSI and SAR images. This network is further enhanced with the addition of newly designed modules to complete the global-local fusion for cloud removal. Of the five methods mentioned above, SpA GAN and SAR2OPT perform single-modal predictions of cloud-free optical images, while the latter three, SAR-Opt-cGAN, DSen2-CR, and GLF-CR, are two-modal fusion methods, both of which are direct fusion.

The quantitative results of our experiments are shown in Table I. As can be seen from the table, the proposed TransFusion-CR outperforms comparative methods across all metrics. We select five scenes for visual result demonstration, as shown in Fig. 4. Further qualitative analysis is conducted based on these visual results. The first column displays Sentinel-2 cloudy inputs, with the first two items being mixed scenes of farmland, roads, and woods. The third scene features a mountainous area, while the last two scenes are grasslands. The five scenes are arranged in ascending order of cloud coverage, in turn, is <10%, 10%∼30%, 30%∼50%, 50%∼70%, and >70%. The second column presents the corresponding Sentinel-1 SAR inputs. The experimental results of the comparative methods are displayed in the third to sixth columns, while the results of our proposed method are shown in the

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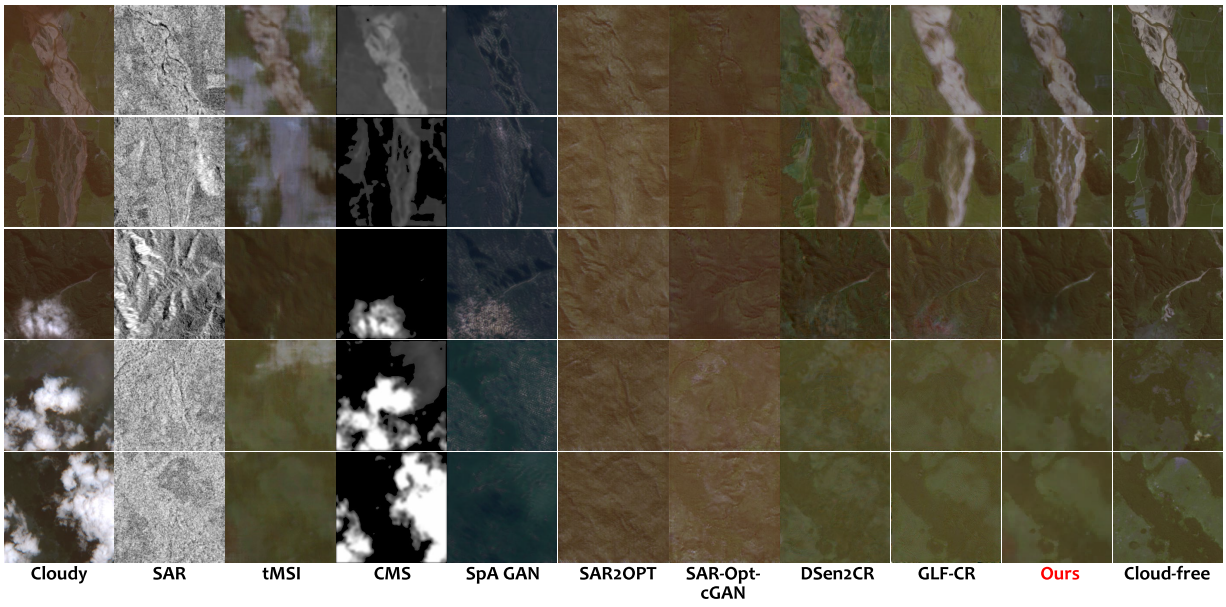

Fig. 4. Qualitative results of the baselines and the proposed method, where tMSI denotes translated MSI. The selected images display results for various land types, including riverbed, mixed landform, mountain, and lawn. The last two rows present optical inputs with a greater than 50% of thick cloud cover, whereas SAR images provide more limited visual information, making it challenging to reconstruct cloud area information. This demonstrates the capabilities of each method in reconstructing texture and spectral details under these conditions.

seventh column. The last column contains cloud-free ground truth images. It is visually evident that our proposed method achieves the best cloud removal results. Detailed analysis is provided as follows.

Among the comparative methods, SpA GAN relies only on a single optical modal for cloud removal, without the supplementary information from SAR. Its results are shown in the third column of Fig. 4. The result images generated by this method have a uniform color tone. In cloud-free areas, the method captures the texture and structure of the ground but falls short in terms of reconstruction fidelity. It is noticeable from the results that SpA GAN is sensitive to shadows to some extent, which can result in the generation of darker patches. In cloudy areas, the method may generate patches that do not correspond to the actual ground details, failing to restore the surface information obscured by thick clouds. This demonstrates the limitations of relying exclusively on a single optical modal for cloud removal.

SAR2OPT and SAR-Opt-cGAN encounter comparable issues in their generated results due to their similar structures. It can be seen that SAR2OPT lacks optical cloudy image input in its final prediction stage, resulting in high consistency with the texture and structure of the SAR images, but discrepancies with ground truth. The spectral information in the results is derived from the mapping relationship learned during training. However, the mapping's accuracy is not sufficient to support more varied terrestrial spectral changes, which can lead to spectral distortion. SAR-Opt-cGAN uses SAR as prior information and also uses cloudy inputs to remove clouds, providing more reliable spectral information. However, due to its similar cGAN backbone network to SAR2OPT, spectral distortion can still occur. Furthermore, the reconstruction of information in areas with thick clouds indicates inadequate SAR information capture, resulting in unsuccessful detail reconstruction and affecting the surface texture in cloud-free areas. DSen2-CR, which incorporates newly designed residual blocks, captures more precise structural information in SAR images, providing additional gains. However, it produces certain artifacts during the image reconstruction and generates cluttered details in heavily cloud-contaminated areas. GLF-CR draws inspiration from the design of the transformer. As can be seen from Fig. 4, it addresses the challenge of generating cluttered details by balancing global and local fusion in the image. However, this approach can result in some blurring. This indicates that in dual-modal fusion for cloud removal, effectively utilizing the information provided by different modal images is crucial, which directly impacts the effectiveness of the cloud removal process.

Our proposed method outperforms visually effective DSen2-CR in all metrics. As shown in Fig. 4, it maintains texture details in cloud-free areas better than DSen2-CR, such as mountain ridges. Especially for cases where the percentage of cloud cover is greater than 50% (the last two rows in Fig. 4), the proposed method generates fewer false details or blurred artifacts compared to DSen2-CR. In cloudy areas, the SFE module extracts richer high- and low-frequency features from SAR images and generates the translated MSI (as shown in the third column in Fig. 4) during the translation phase. In contrast to the SAR image, which only contains the structural information, the translated MSI generates and supplements additional spectral bands of optical color information while maintaining the SAR structural information. The translated MSI is sent into the next fusion phase for subsequent weighted fusion, as well as the final reconstruction of the surface texture and color details. The proposed method can realize superior cloud removal results to DSen2-CR. In contrast to the SAR2OPT

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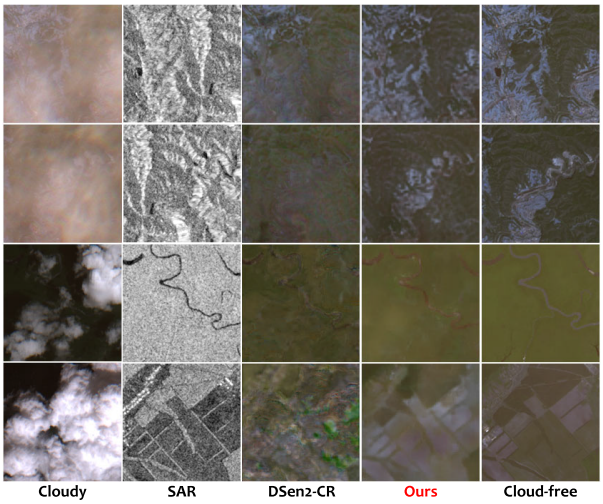

Fig. 5. Qualitative experimental results of DSen2-CR and TransFusion-CR (Ours) on DatasetV2. The images from the first to the fifth column represent the cloudy, SAR, DSen2-CR, ours, and cloud-free images, respectively.

and SAR-Opt-cGAN methods, our approach assigns varying information learning weights to different areas based on cloud cover. This approach not only maintains detail accuracy but also preserves true spectral information to a greater extent. These significant improvements demonstrate that our method effectively utilizes and optimizes the spectral information provided by SAR in image fusion, enhancing the effects of detail preservation and reconstruction.

D. Comparison on Different Datasets

To further test the superiority of our proposed model, we select data from SEN12MS-CR to create a more complex dataset for training and testing, which includes a wider range of terrain types (such as farmland, roads, rivers, lakes, and urban areas), as illustrated in Fig. 5. The percentage of cloud cover for all MSI images in DatasetV2 is above 70%. The cloud removal challenge is significantly greater with this dataset.

DSen2-CR, which performs well in the comparative analysis in Section IV-C, is chosen as the baseline for the experiments on DatasetV2. The quantitative results are presented in Table II. The results indicate that in datasets with more complex details and a higher percentage of cloud cover, the performance of both methods decreases to some extent. However, the decline is generally smaller for the proposed TransFusion-CR and it outperforms DSen2-CR across all three metrics, which demonstrates the superiority of our method.

E. Cloud Removal on Large Scene

To verify the cloud-removal effectiveness of the proposed method in a large scene, we choose an image of 1024 × 1792 px^2 as the test image. Similar to the experiments on DatasetV2 in Section IV-D, DSen2-CR is chosen as the comparison method. The experimental results are shown in Fig. 6. It can be observed that both methods are capable of removing clouds, however, the reconstruction quality of ground information in large scenes inevitably declines due to the sharp increase in the richness of detailed features. In contrast, the reconstructed color of the proposed method is more closely aligned with the ground truth, and the reconstruction quality in certain details is superior to that of DSen2-CR.

TABLE II
QUANTITATIVE RESULTS OF DSEN2-CR AND TRANSFUSION-CR ON DIFFERENT DATASETS. Δ DENOTES THE VARIATION IN THE RESULTS OF THE METHOD ON THE DATASETV1 AND DATASETV2

Method	Dataset	PSNR (dB) ↑	SSIM ↑	SAM (°) ↓
DSen2-CR [23]	DatasetV1	25.2412	0.7886	5.2650
	DatasetV2	21.6854	0.6920	7.1200
		Δ%14.09	*Δ%12.25*	*Δ%35.23*
TransFusion-CR (Ours)	DatasetV1	25.7649	0.8344	4.4000
	DatasetV2	**22.9333**	**0.7764**	**6.0060**
		Δ%10.99	*Δ%6.95*	*Δ%36.50*

F. Ablation Study

To validate the effectiveness and importance of each component in the TransFusion-CR, the ablation experiments are conducted on DatasetV1. The settings for learning rate and other parameters in these experiments are consistent with those in Section IV-B. In the ablation studies, we sequentially remove the guidance of CMS, the SFE module, the CFF module, and the $\mathcal{L}_{\text{loss}}$ loss from the network. The final quantitative results (as shown in Table III) demonstrate the effectiveness of each design element proposed. We select four scenes for visual display (as shown in Fig. 7), followed by a more detailed analysis and discussion:

1) Analysis of the Guidance of CMS: The design of CMS primarily guides feature fusion in the CFF module, to enhance the effectiveness of the fusion process. This is achieved by guiding the learning of the translated MSI toward the thick cloud region while directing the learning of the MSI input toward the cloud-free or thin cloud region. This approach allows for the learning effect with weights to be achieved. In this experiment, the two modules of SFE and CFF in the network are retained, but the CMS bootstrapping in the CFF module is removed, and fusion is performed directly. The results are labeled with "*w/o* CMS." It can be observed that the removal of CMS results in a decrease in PSNR of the predicted image, despite an increase in the other two metrics. The image displays brighter regions that deviate from the true value, and there are instances of minor detail reconstruction failures, particularly in thick cloud areas. Consequently, these observations suggest that CMS can enhance the fusion process.

2) Analysis of SFE and CFF Module: The SFE module is designed to make the most of the information from SAR images, while the CFF module is proposed to enhance the network's ability to learn from both cloud-free and cloudy MSI images. This is achieved by allocating learning weights to each image with varying cloudy conditions/cloud cover. The ultimate goal is to improve the prediction of spatial details and spectral information. We conduct three experiments on DatasetV1: i) removing the SFE module from each layer of the network's translation phase, which reduces the frequency

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TABLE III

QUANTITATIVE ABLATION STUDY RESULTS OF TRANSFUSION-CR NETWORK, WITH OR WITHOUT (*w/o*) SFE MODULE, CMS-GUIDED FEATURE FUSION (CFF) MODULE, AND $\mathcal{L}_{\text{loss}}$ LOSS. THE RESULTS DEMONSTRATE THE EFFECTIVENESS OF EACH ELEMENT IN THE PROPOSED NETWORK FOR CLOUD REMOVAL

Method	**Guidance**	**Module**		**Loss**		**PSNR** (dB) ↑	**SSIM** ↑	**SAM** (°) ↓
	CMS	SFE	CFF	$L1$	$\mathcal{L}_{loss}$			
w/o CMS		✓	✓		✓	25.5392	0.8380	4.3804
w/o SFE	✓		✓		✓	25.5439	0.8266	4.6370
w/o CFF		✓			✓	25.1517	0.8184	4.6750
w/o SFE & CFF					✓	25.1472	0.8145	4.9340
w/o SFE & CFF & $\mathcal{L}_{loss}$				✓		24.9297	0.8109	4.9672
TransFusion-CR (ours)	✓	✓	✓		✓	**25.7649**	**0.8344**	**4.4000**

Fig. 6. Qualitative experimental results on large scenes for cloud removal. The experiment is conducted on the image of size 1024 × 1792 px^2.

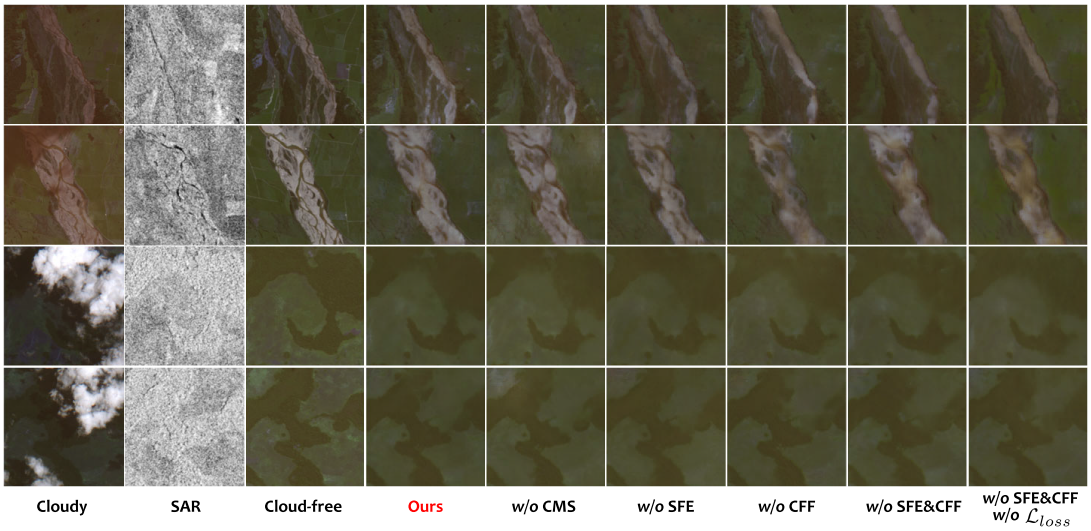

Fig. 7. Qualitative ablation study results, with column information arranged as follows from left to right: cloudy, SAR, cloud-free (ground truth), TransFusion-CR (ours), *w/o* CMS, *w/o* SFE, *w/o* CFF, *w/o* SFE & CFF, and *w/o* SFE & CFF & $\mathcal{L}_{\text{loss}}$. These results correspond to the information presented in Table III.

and spectral information extracted from SAR; ii) removing the CFF module from the network's fusion phase, resulting in a direct concatenation and fusion of inputs without guiding the learning direction based on cloud areas; and iii) removing both SFE and CFF modules at the same time, leaving only the $\mathcal{L}_{\text{loss}}$ loss design. The top three rows of Table III show a decrease in performance metrics for all three methods, i.e., *w/o* SFE, *w/o* CFF, and *w/o* SFE & CFF. The removal of the CFF module results in a more significant decrease. In addition, the experimental results of the methods without these modules display more blurred surface details and small-scale inaccuracies in spectral reconstruction, as observed from the fifth to the seventh columns in Fig. 7.

3) Analysis of $\mathcal{L}_{loss}$: A new loss, $\mathcal{L}_{\text{loss}}$, is developed to balance the network's learning focus between the first translation phase and the second fusion phase. To verify its effectiveness, we replace it with a vanilla $L1$ loss to train the network. The qualitative comparison results are presented in the rows labeled with "*w/o* SFE & CFF" and "*w/o* SFE & CFF & $\mathcal{L}_{\text{loss}}$," corresponding to the last two rows in Fig. 7. The experimental results demonstrate that the newly designed loss achieves more real reconstruction of spectral information and reduces blurriness in detail such as the edges of lawns and mountain textures, which suggests that the loss function effectively optimizes the network training toward the intended goals.

G. Efficiency Comparison

In Table IV, we present the quantitative results of inference time, frames per second (FPS), floating point operations (FLOPs), and parameter count (Params) between the proposed TransFusion-CR method and the comparison baselines. The

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TABLE IV
QUANTITATIVE COMPARISON OF THE INFERENCE TIME, FPS, FLOPS, AND PARAMETERS OF EACH METHOD

Method	Inference Time		FPS	FLOPs (G)	Params (M)
	Mean (ms)	Std (ms)			
SpA GAN [16]	16.316	0.883	61.29	18.82	2.98
SAR2OPT [9]	20.783	1.389	48.12	23.51	10.78
SAR-Opt-cGAN [22]	22.466	0.992	44.51	27.59	12.63
DSen2-CR [23]	137.937	9.927	7.25	483.77	18.94
GLF-CR [7]	195.368	8.776	5.12	95.13	14.68
TransFusion-CR (ours)	25.314	3.412	39.50	31.39	43.96

inference time considers both the mean time (Mean) and standard deviation (Std) of 300 repeated calculations on a NVIDIA RTX 3090Ti GPU with a batch size of 1. The comparison baselines employ the direct fusion strategy for cloud removal, whereas the proposed TransFusion-CR method is a two-phase fusion strategy comprising two subnetworks responsible for translation and fusion, respectively. Consequently, the parameter count is higher than that of the direct fusion methods with a single phase. It is obvious from Table IV that while TransFusion-CR exhibits a higher parameter count, it has a noticeable advantage in inference time and lower model computation, particularly in comparison to DSen2-CR and GLF-CR.

V. DISCUSSION AND CONCLUSION

A. Challenging Conditions

We use two datasets for experiments, i.e., DatasetV1 and DatasetV2; both are constructed from images in SEN12MS-CR. The primary distinction between the datasets is the variety of terrain types and the richness level of detail and surface information. DatasetV1 has fewer terrain types than DatasetV2. Consequently, it demonstrates better visual effects and higher evaluation metrics (see Section IV-D). Under extreme conditions, particularly when thick clouds completely obscure the surface, our method experiences limitations. This mainly occurs in incomplete spectral recovery (as shown in the last row of Fig. 5, where land is segmented accurately based on SAR but partial spectral distortion occurs). This issue arises because optical images do not provide any spectral information or spatial detail reference under such conditions, leaving SAR as the only source of prior information. However, SAR lacks enough spectral priors, which may lead to incorrect spectral generation. The proposed SFE module has been proved in Section IV-F to be able to extract and expand the information of SAR for cloud removal and boost gains. Consequently, this issue should be better addressed in larger datasets or under conditions with restricted terrain types.

B. Conclusion

In this article, we introduce TransFusion-CR, a novel end-to-end two-phase deep neural network that utilizes both the MSI and SAR images. Notably, it can effectively remove clouds and reconstruct ground information in cloudy areas, while preserving the integrity of information in cloud-free regions. To fully utilize SAR for reconstructing ground details, and maintain global spectral and spatial details, we adopt a two-phase approach in the algorithm. In the translation phase, SAR is translated into a synthetic cloud-free MSI. Its spatial features are converted to approximate optical image features and spectral dimension information is generated. During this process, the SFE module is established to process different frequency information of SAR separately, so that the translated MSI possesses clear and reliable spatial–spectral information. Subsequently, feature learning and image fusion between the synthetic MSI and the cloudy MSI are conducted in the fusion phase. To achieve more efficient and accurate fusion across varying spectral and spatial regions, the CFF module fuses the features from triple sources in cloudy and cloud-free areas with different weights. It then uses the cloud feature map as guidance and ultimately generates the predicted cloud-free MSI after the decoding process of features. Through the guided-fusion approach, the areas previously covered by clouds are reconstructed with accurate detail and structure, and spectral distortion is corrected to some extent; while the previous cloud-free areas still remain clear and uncorrupted after the reconstruction. Finally, to enhance the cloud removal effectiveness of the network, a new loss function is designed. It balances the network's learning focus between the two phases, reducing detail blur and improving reconstruction accuracy. In the future, we will further optimize the network layer design and refine the training scheme to enhance the network's generalization capabilities. This attempt will allow the method to improve training efficiency and achieve more reliable cloud removal results.

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